

Practical Week – 2

Task : 1

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>

#define MAX_INPUT 100

int main()
{
    char input[MAX_INPUT];
    char *args[MAX_INPUT];
    int i = 0;
    pid_t pid;
    int status;

    printf("Enter a Linux command: ");
    fgets(input, sizeof(input), stdin);

    // Remove newline
    input[strcspn(input, "\n")] = '\0';

    // Convert input into arguments
    char *token = strtok(input, " ");

    while (token != NULL)
    {
```

```
    args[i++] = token;
    token = strtok(NULL, " ");
}

args[i] = NULL;

// Create child process
pid = fork();

if (pid < 0)
{
    perror("fork failed");
    return 1;
}

if (pid == 0)
{
    // Child process
    printf("\nChild Process\n");
    printf("Parent PID : %d\n", getppid());
    printf("Child PID : %d\n", getpid());

    printf("\nExecuting command: %s\n", args[0]);

    // Execute command
    execvp(args[0], args);

    // Only reached if execvp fails
    perror("execvp failed");
    exit(1);
}
```

```

}
else
{
    // Parent process

    printf("\nParent Process\n");
    printf("Parent PID : %d\n", getpid());
    printf("Child PID : %d\n", pid);


    // Wait for child
    wait(&status);


    printf("\nChild process completed.\n");
}

return 0;
}

```

Output :

```

avulasanjana@SanjanaReddy:~$ nano command_executor.c
avulasanjana@SanjanaReddy:~$ gcc command_executor.c -o command_executor
avulasanjana@SanjanaReddy:~$ ./command_executor
Enter a Linux command: ls

Parent Process
Parent PID : 492
Child PID : 493

Child Process
Parent PID : 492
Child PID : 493

Executing command: ls
File1.txt      command_executor      keyboard_input      parser_quotes.c
File2.txt      command_executor.c    keyboard_input.c    process
LICENSE        docs                  main_loop           process.c
Makefile       file_copy             main_loop.c         process.c.save
OSSP           file_copy.c           obj                 scripts
README.md      fork_exec             parser_double_quotes src
assets         fork_exec.c           parser_double_quotes.c tests
bin            include               parser_quotes
Child process completed.

```

Task : 2

```
avulasanjana@SanjanaReddy:~$ uname -a
Linux SanjanaReddy 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC
Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
avulasanjana@SanjanaReddy:~$ lscpu | head -20
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      16
On-line CPU(s) list:        0-15
Vendor ID:                   GenuineIntel
Model name:                  13th Gen Intel(R) Core(TM) i7-13620
H
CPU family:                  6
Model:                      186
Thread(s) per core:         2
Core(s) per socket:         8
Socket(s):                   1
Stepping:                    2
BogoMIPS:                    5836.80
Flags:                       fpu vme de pse tsc msr pae mce cx8
apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall
nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable nons
top_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 ss
e4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hyperv
isor lahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow
ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed
adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnm
i umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_clear se
rialize ibt flush_lld arch_capabilities
Virtualization:              VT-x
Hypervisor vendor:           Microsoft
Virtualization type:         full
Lld cache:                   384 KiB (8 instances)
avulasanjana@SanjanaReddy:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root          1        0  0 06:57 ?        00:00:00 /sbin/init
root          2        1  0 06:57 hvcd0    00:00:00 /init
root          6        2  0 06:57 hvcd0    00:00:00 plan9 --control-socket 7
root         46        1  0 06:57 ?        00:00:00 /usr/lib/systemd/systemd
systemd+     81        1  0 06:57 ?        00:00:00 /usr/lib/systemd/systemd
root         88        1  0 06:57 ?        00:00:00 /usr/lib/systemd/systemd
root        107        1  0 06:57 ?        00:00:00 /bin/sh /usr/lib/systemd
root        108        1  0 06:57 ?        00:00:00 /usr/sbin/cron -f -P
message+    109        1  0 06:57 ?        00:00:00 @dbus-daemon --system --
root       128        1  0 06:57 ?        00:00:00 /usr/bin/python3 /usr/bi
```

```

root      319      316  0 06:57 ?          00:00:00 /init
avulasa+  323      319  0 06:57 pts/0      00:00:00 -bash
root      326        2  0 06:57 ?          00:00:00 login -- avulasanjana
avulasa+  379        1  0 06:57 ?          00:00:00 /usr/lib/systemd/systemd
avulasa+  381      379  0 06:57 ?          00:00:00 (sd-pam)
avulasa+  407      326  0 06:57 pts/1      00:00:00 -bash
root      570        2  0 07:01 ?          00:00:00 /init
root      571      570  0 07:01 ?          00:00:00 /init
avulasa+  573      571  0 07:01 pts/2      00:00:00 -bash
avulasa+  625      573  0 07:02 pts/2      00:00:00 ps -ef

```

avulasanjana@SanjanaReddy:~\$ top

```

top - 07:02:44 up 5 min,  1 user,  load average: 0.00, 0.02, 0.00
Tasks: 28 total,  1 running, 27 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.
MiB Mem :  7771.7 total,  6355.2 free,   572.5 used,  996.6 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used.  7199.1 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+
637	avulasa+	20	0	8184	6132	4020	R	0.3	0.1	0:00.01
1	root	20	0	24332	15404	11584	S	0.0	0.2	0:00.82
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.01
6	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00
46	root	19	-1	50428	16584	15312	S	0.0	0.2	0:00.16
81	systemd+	20	0	22540	14588	11968	S	0.0	0.2	0:00.07
88	root	20	0	35408	12364	9228	S	0.0	0.2	0:00.35
107	root	20	0	2888	1952	1820	S	0.0	0.0	0:00.04
108	root	20	0	4472	3132	2884	S	0.0	0.0	0:00.00
109	message+	20	0	8820	5436	4780	S	0.0	0.1	0:00.05
128	root	20	0	44092	29804	14664	S	0.0	0.4	0:00.23
155	root	20	0	18420	9372	8140	S	0.0	0.1	0:00.08
167	root	20	0	1792044	15276	12616	S	0.0	0.2	0:00.10
192	syslog	20	0	220548	5952	4656	S	0.0	0.1	0:00.07
208	root	20	0	5492	2740	2552	S	0.0	0.0	0:00.01
213	_chrony	20	0	20420	7864	7032	S	0.0	0.1	0:00.03
227	_chrony	20	0	12092	2440	1796	S	0.0	0.0	0:00.00
229	root	20	0	123456	32572	17960	S	0.0	0.4	0:00.19
316	root	20	0	3184	1116	980	S	0.0	0.0	0:00.00
319	root	20	0	3200	1252	1108	S	0.0	0.0	0:00.30
323	avulasa+	20	0	6268	5528	3860	S	0.0	0.1	0:00.05
326	root	20	0	8652	5540	4692	S	0.0	0.1	0:00.01
379	avulasa+	20	0	22332	13536	10928	S	0.0	0.2	0:00.13
381	avulasa+	20	0	22912	4092	2088	S	0.0	0.1	0:00.00
407	avulasa+	20	0	6136	5492	3928	S	0.0	0.1	0:00.03
570	root	20	0	3184	1124	980	S	0.0	0.0	0:00.00
571	root	20	0	3200	1260	1108	S	0.0	0.0	0:00.00
573	avulasa+	20	0	6268	5536	3864	S	0.0	0.1	0:00.05